

7.2 Analysing variables one at a time [↕]

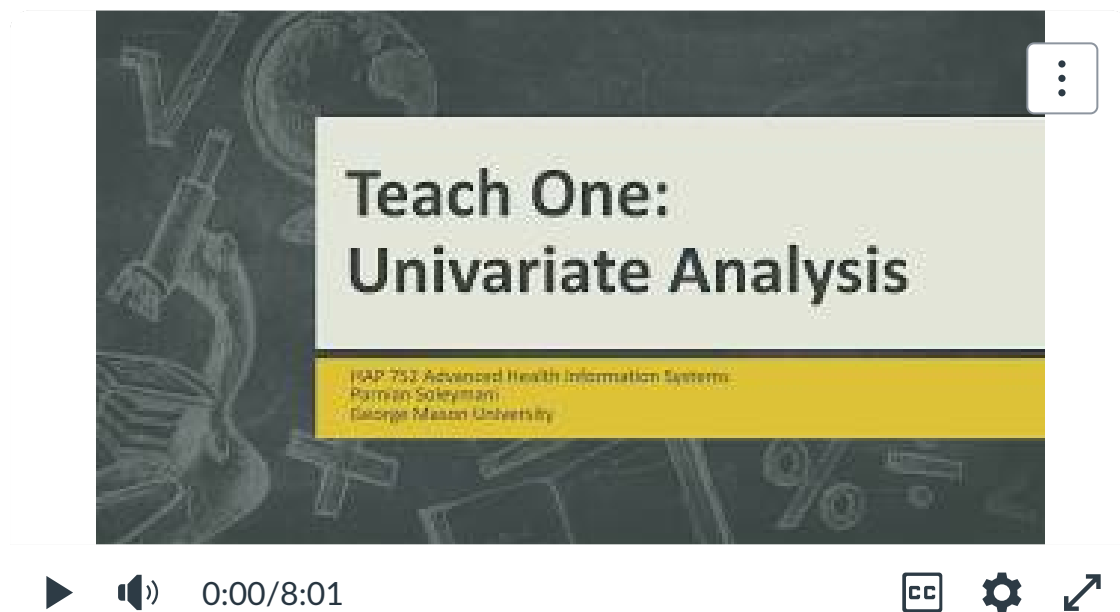
In general, it is advisable to initially examine one variable (column) at a time. This is called *univariate analysis*, and it focuses on the distribution of one variable's values.

□ Watch: Introduction to univariate analysis [8.00 mins]

This video provides an overview of univariate analysis. Different ways of conducting univariate analysis are discussed and two specific types of variables are examined.

As you watch, consider:

- How univariate analysis can be used in exploratory data analysis.
- Why univariate analysis should be used during the process of exploratory data analysis.



Source: © LadyP (2018).

Pivot tables and pivot charts are useful tools for exploring data.

- A *pivot table* calculates, summarises and analyses data to show comparisons patterns and trends of one or more variables.
- A *pivot chart* provides an interactive visualisation of data.

□ Watch: Excel pivot tables explained [13.21 mins]

This video demonstrates how simple it is to get data insights quickly by creating a pivot table using MS Excel. Filters and slicers are also explained.

As you watch, consider:

- How could pivot tables and slicers be used at various stages of investigating and telling the story held within a dataset?
- What advantages and disadvantages might there be when using slicers in professional data visualisations?



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As discussed in [Module 6 \(https://learn.usc.edu.au/courses/35242/modules/items/924165\)](https://learn.usc.edu.au/courses/35242/modules/items/924165), *box and whisker* charts are a useful way to visualise the distribution of a quantitative variable. They present information about a variable's central tendency, data points with extreme values and the symmetry in a distribution.

When *outliers* are present in a box and whisker chart, it is a good idea to investigate the related records. Outliers may indicate an inaccurate value in the data. If the investigation reveals that the data value is erroneous or there were issues in how it was observed, including it in the analysis may be inappropriate and therefore it may be appropriate to omit it. However, be careful removing outliers without good cause, as it can distort the analysis because the variable's variability is reduced.